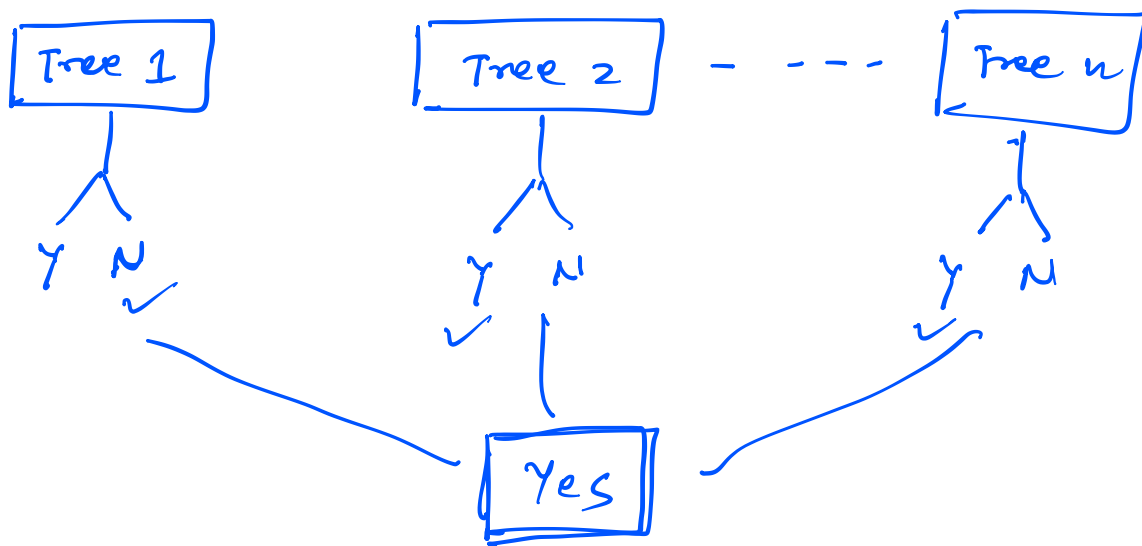


Random Forest

One decision tree has high variance. - slight change in data can cause change in rules (decisions).

Instead of training one tree, train multiple trees on different subset of data. → Ensemble.



* Bagging. :- Bootstrap Aggregating.
variance reduction Strategy.

B trees.

$$\hat{f}(x) = \frac{1}{B} \sum_{b=1}^B f_b(x) \quad \text{for regression.}$$

$$\hat{f}(x) = \text{mode } f_b(x) \quad \text{for classification}$$

$$\text{Total error} = \text{bias}^2 + \text{variance} + \text{noise}$$

* Hard vote: - Majority voting.

* Soft vote: - Average of the probabilities.

Average for regression problems.

Ingredient:

- ① Each strong individual tree $\rightarrow$ high variance
- ② Trained on diff. subset of data.
- ③ Combining gives you a great model.

Row wise randomness $\rightarrow$ samples
column wise randomness $\rightarrow$ features.

① Bootstrapping:

A, B, C, D $\rightarrow$ original dataset.

sampling with replacement.

① C, A, C, D

② B, B, C, A

In n draws, what is the chance that one particular row is never selected?

Total n samples.

uniform distribution :- $(\frac{1}{n})$ is the prob. of being selected,

$(1 - \frac{1}{n}) \Rightarrow$ not being selected,

$(1 - \frac{1}{n})^n \rightarrow$ row not being selected at all.
Dataset is huge $n \rightarrow \infty$
 $\rightarrow e^{-1} \approx 0.37$

with 37% probab. a row can't be a part of random subset,

O O B : $\sim 37\%$ of data is not part of your bootstrapped dataset.

* feature randomness:

sample $\sqrt{\text{\# features}}$ for each tree.

* Bagging recipe:

- ① Do bootstrapping (sample rows)
- ② Do feature selection (random)
- ③ Train a base learner B
- ④ Aggregate their Prediction.

* Strength vs. correlation:

$m \rightarrow \#$ of features.

- ① if m is low, correlation is also low.
but each tree is a weak learner.
- ② if m is high, correlation is also high,
but each tree is a strong learner.

selecting no. of features.

- ① $\sqrt{m}$
- ② 30%
- ③ $\log_2(m)$

* can RF extrapolate: No

